

Gitlab Usage Guide

What is Git?

Git is a version control system used to track changes in computer files. Git's primary purpose is to manage any changes made in one or more projects over a given period of time. It helps coordinate work among members of a project team and tracks progress over time.

What is Gitlab?

GitLab is a web-based DevOps lifecycle tool that provides a Git-repository manager providing wiki, issue-tracking and continuous integration and deployment pipeline features, using an open-source license, developed by GitLab Inc.

How to work with GitLab?

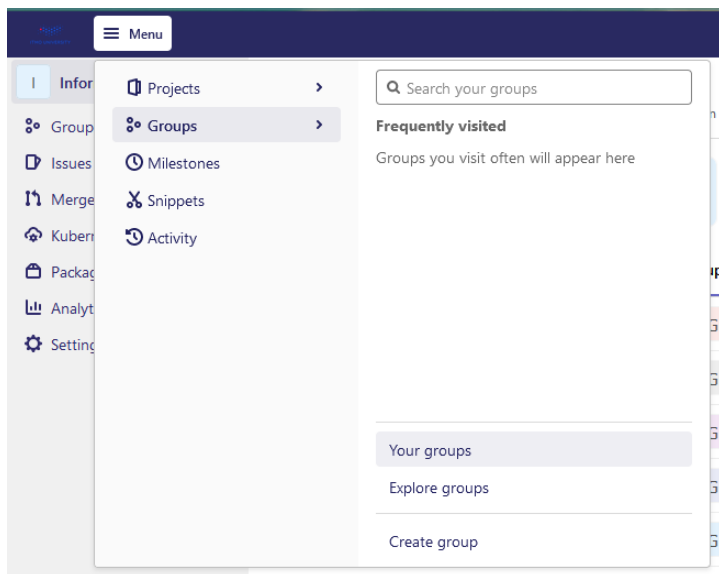
You will get access to the subgroup and project.

The discipline's group names «Information Systems (HDU)» and private. So, only group members can see it. Here is a subgroup «groupN», where N is your group number. Each subgroup is private also, so only its members can access projects of subgroup. More about groups, subgroups and members at https://docs.gitlab.com/ee/topics/set_up_organization.html and https://docs.gitlab.com/ee/user/project/code_owners.html.

To view groups:

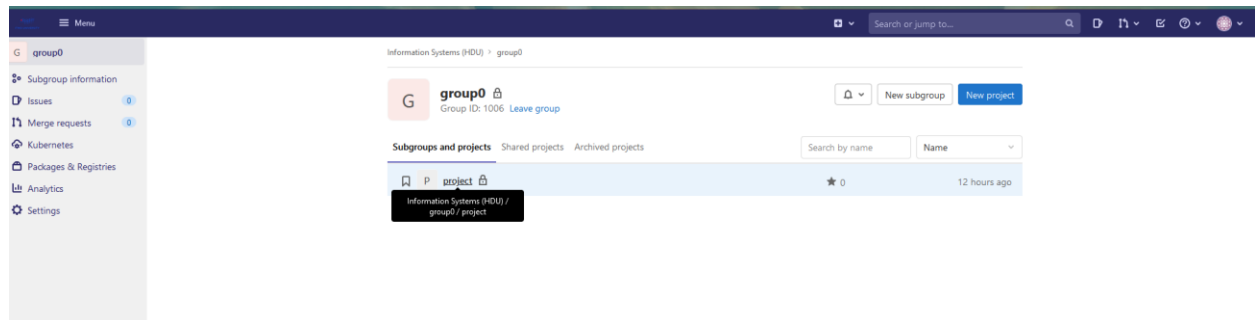
1. On the top bar, select Menu > Groups.
2. Select Your Groups. All groups you are a member of are displayed.

Also the same actions to see your projects.

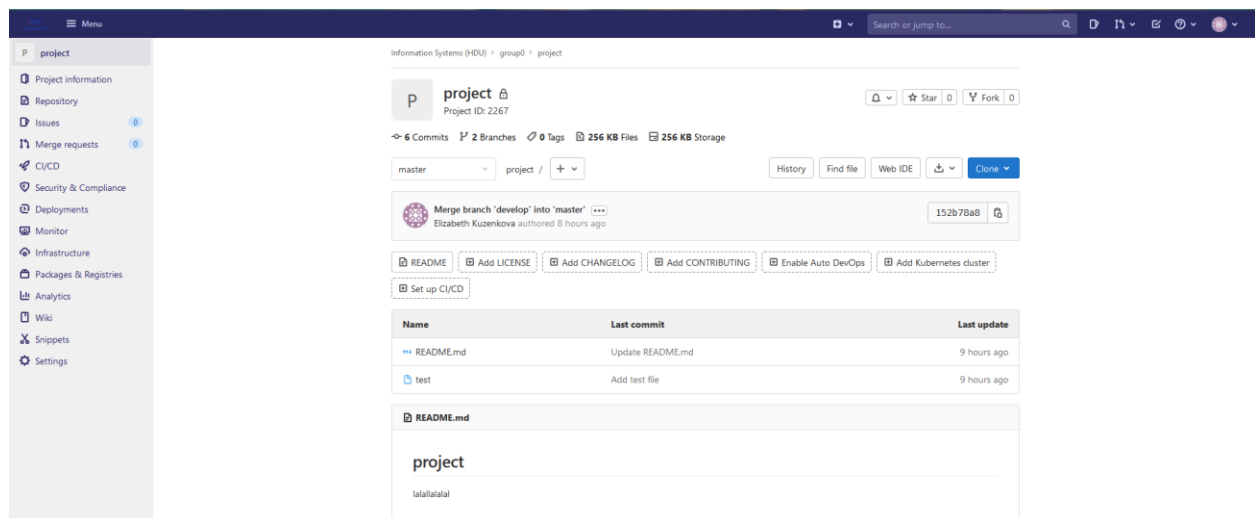


When you open the page of your group, you can see projects (repositories) of this group or subgroup.

More about projects at https://docs.gitlab.com/ee/user/project/working_with_projects.html and at <https://docs.gitlab.com/ee/user/project/repository/>.



For example, here is a repository «project» in subgroup «group0».



It is your repository to publish project tasks.

There is a branch switching control in the upper-left area of the project's webpage. Each branch can be treated as an isolated state of the project sources intended for a particular development goal.

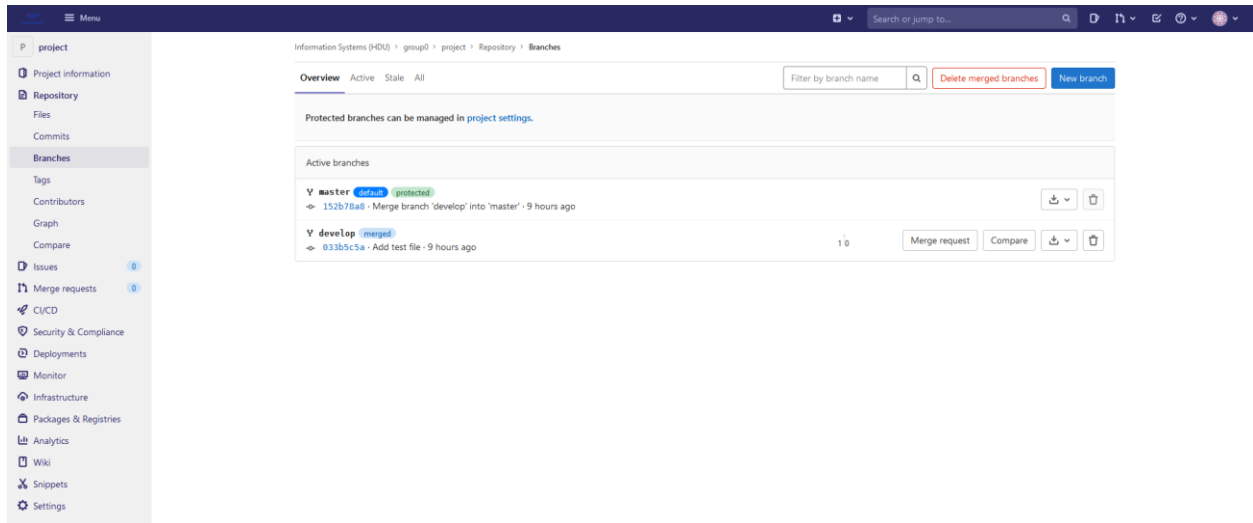
A branch is a version of a project's working tree. You create a branch for each set of related changes you make. This keeps each set of changes separate from each other, allowing changes to be made in parallel, without affecting each other.

For example, "master" branch typically contains production-ready state of the sources corresponding the accomplished version of the product. "develop" branch typically contains shared work-in-progress state of the sources. Other branches can be created to isolate parts of work on the projects while being in process before they'll be ready to merge into "develop" branch to be shared with other participants of the project.

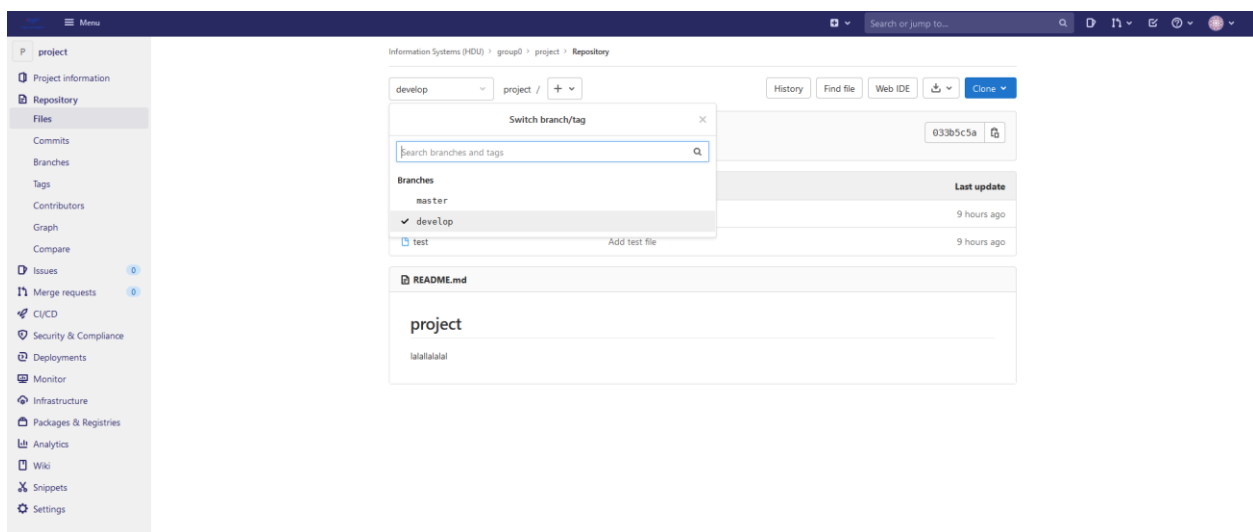
Merge operations and pull-requests are used to manage and transfer modifications between branches.

More about branches at <https://docs.gitlab.com/ee/user/project/repository/branches/>.

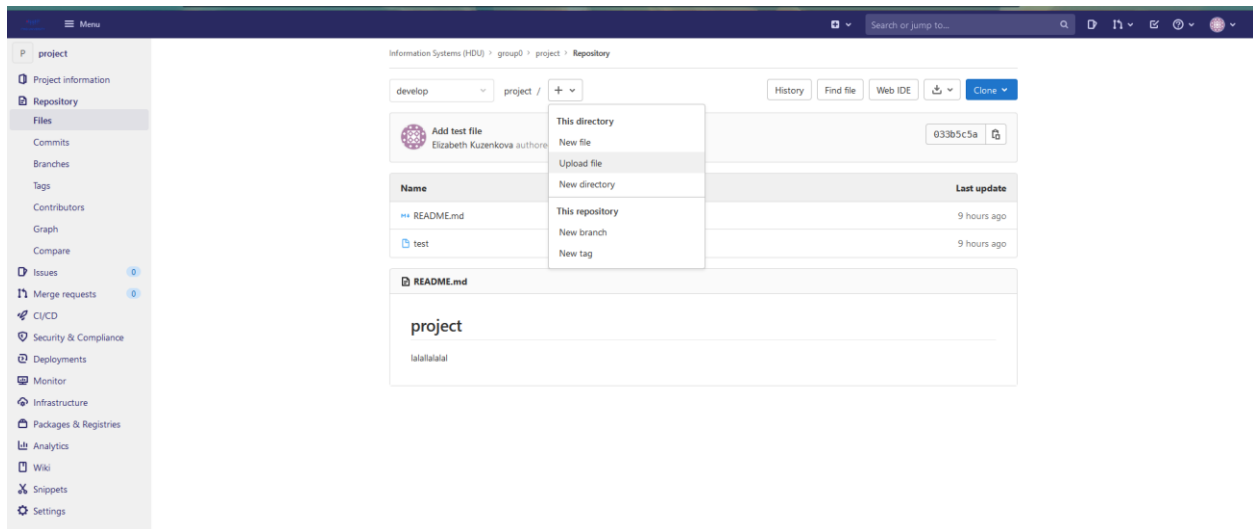
There are 2 branches here.



You get access only to *develop* branch and can create new branches. So, to see actual state of your repository you should switch branch as shown in a next picture.

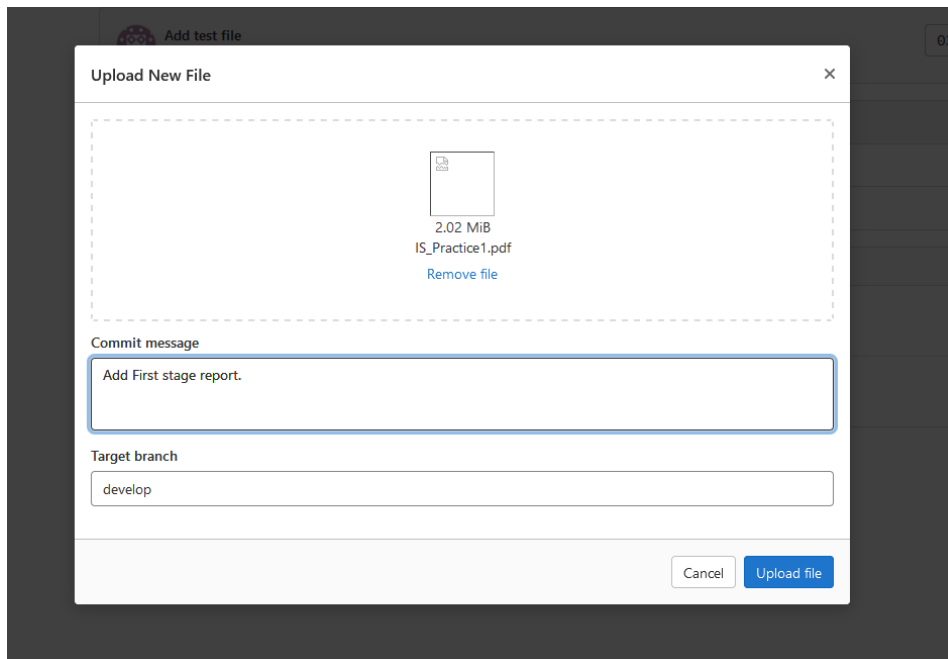


To submit a file to the repository, you should press «+» button near the branches switcher and press «Upload file».

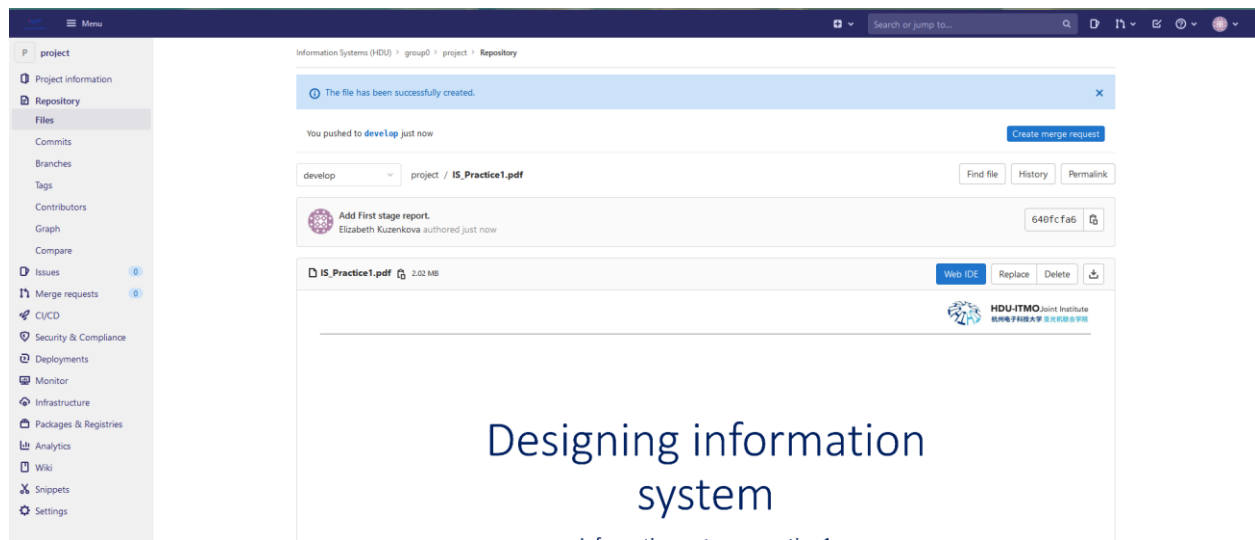


Then such a pop-up window will appear. You should choose the file you want to upload and change «Commit message».

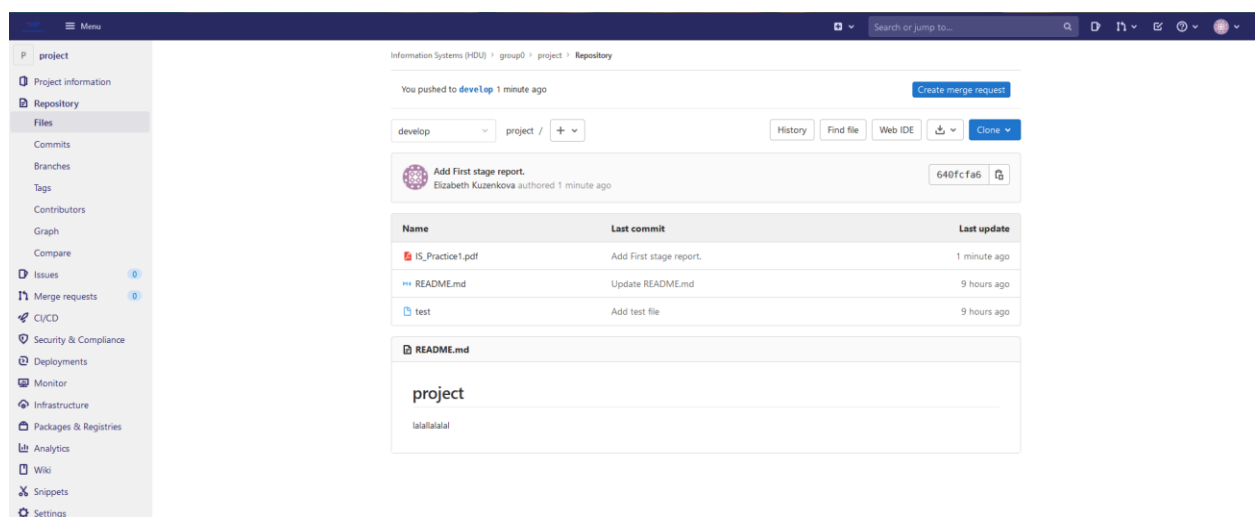
A commit is information about modified files.



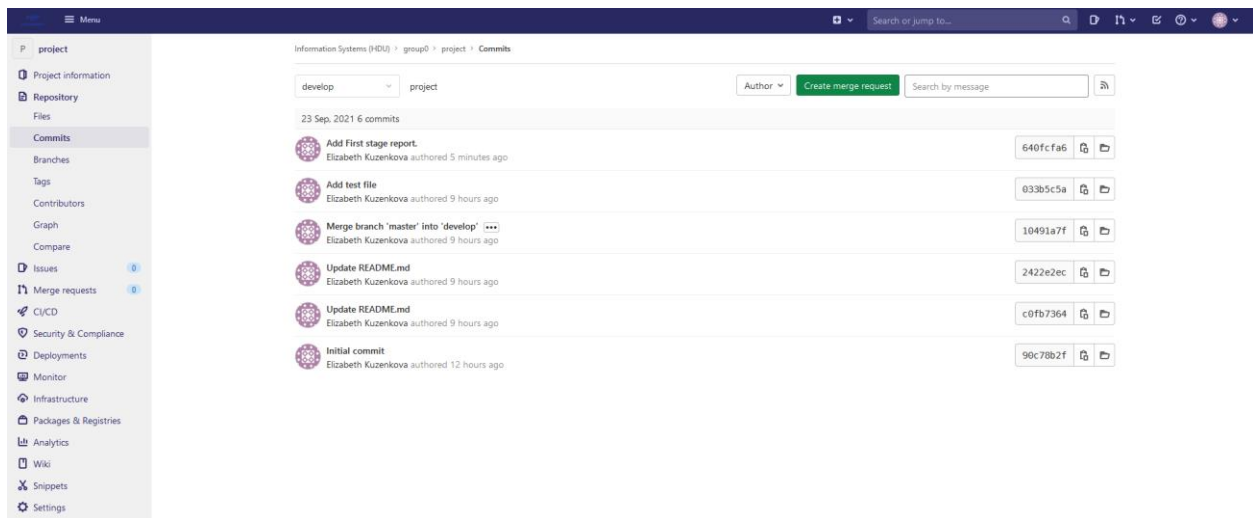
After that you can press «Upload file» button and uploaded file in the repository will be opened. To return to the repository view you can press at the project's name near the branches switcher («project» here) or in a left menu press «Repository» -> «Files» button.



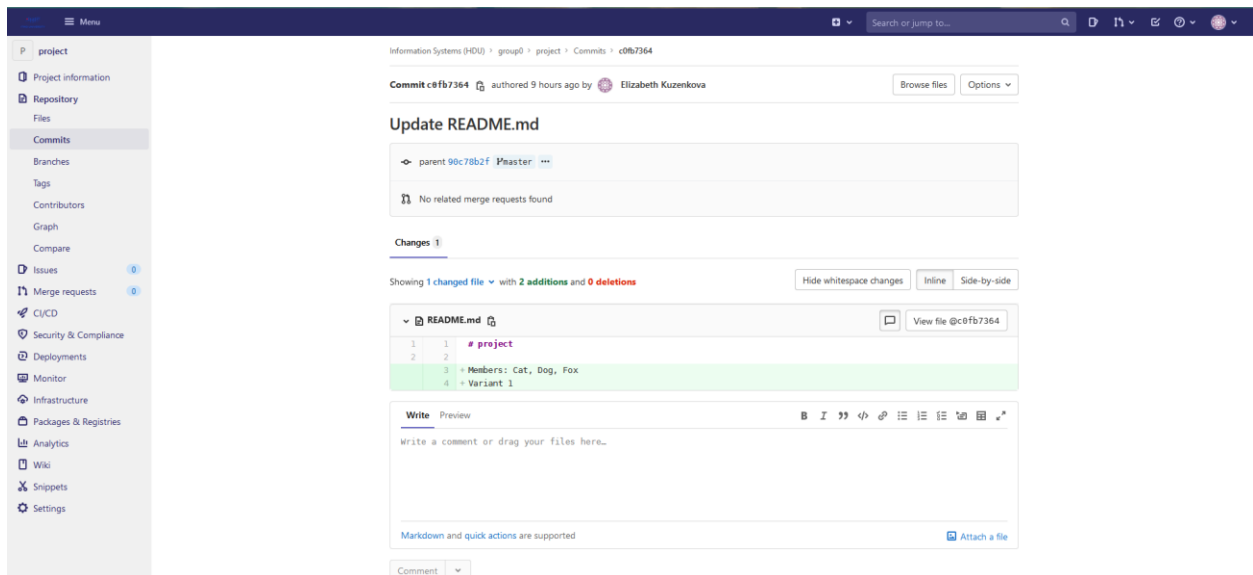
Returning to the repository view you will see that new file was added. Also information about last commit with a commit message and author is available in this view.



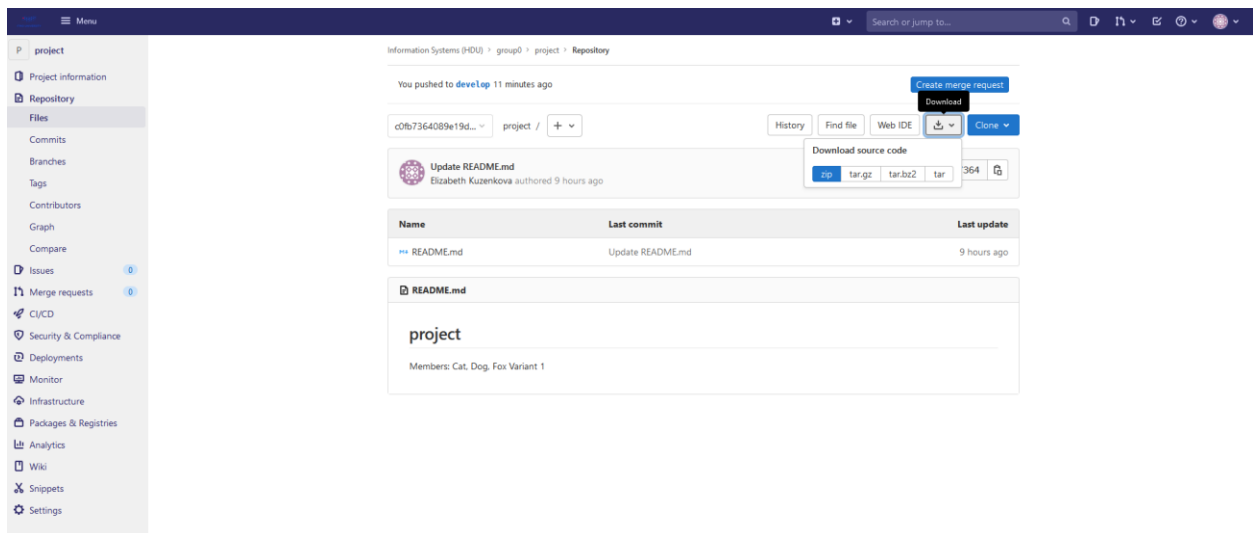
To see all commits you can press «Repository» -> «Commits» button in a left menu. After that you will see a list of commits.



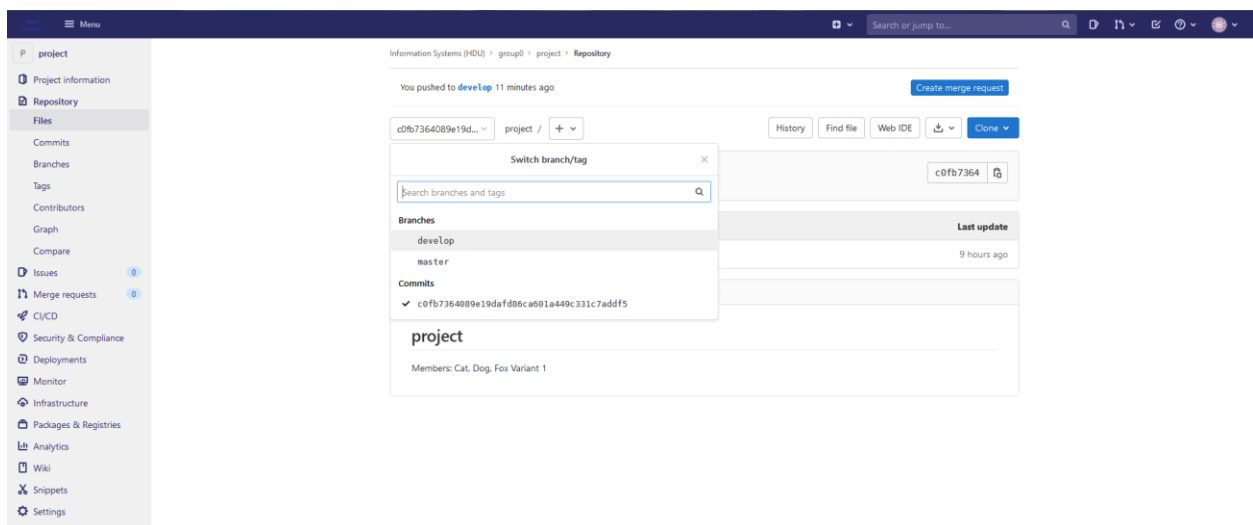
If you press on some commit in this list, you will see changes, which that commit made.



To see the repository state after that commit, press «Browse files» button. You can download whole repository in archive if you press Download button and choose format suitable for you.



In a branch switcher here you see commit identifier. When you upload all necessary files and want the teacher to check your work, press «Create merge request» button.



When you uploaded all necessary files and want the teacher to check your work, press «Create merge request» button.

Merge requests are the way you check source code changes into a branch. More about merge requests at https://docs.gitlab.com/ee/user/project/merge_requests/.

It's important to assign your merge request to me (@E1izabeth), so I could see your merge request and be able to Merge it in a master branch. Choose in «Assignee» @E1izabeth in a list.

Information Systems (HDU) > group0 > project > Merge requests > New

New merge request

From `develop` into `master` [Change branches](#)

Title

Start the title with `Draft:` to prevent a merge request that is a work in progress from being merged before it's ready.
Add description templates to help your contributors communicate effectively!

Description Write Preview

Describe the goal of the changes and what reviewers should be aware of.

Markdown and quick actions are supported [Attach a file](#)

Assignee

Reviewer

Milestone

Labels

Merge options

☐ Delete source branch when merge request is accepted.

☐ Squash commits when merge request is accepted. [?](#)

[Create merge request](#) [Cancel](#)

Now you see your merge request. Here could be some comments about your work from teacher. It could be a comment or a Thread as in the next picture. A Thread is a comment that should be answered by you. You will see comments and threads and be able to write your comments here to communicate with teacher.

About merge requests reviewing https://docs.gitlab.com/ee/user/project/merge_requests/reviews/

Information Systems (HDU) > group0 > project > Merge requests > 12

[Open](#) Created just now by [Elizabeth Kuzenkova](#) [Owner](#) [Edit](#) [Mark as draft](#)

Add First stage report.

Overview [Commits](#) [Changes](#) 1 unresolved thread

Request to merge `develop` into `master` [Open in Web IDE](#) [Check out branch](#) [Download](#)

The source branch is 1 commit behind the target branch

[Approve](#) Approval is optional

[Merge](#) ☐ Delete source branch

> 1 commit and 1 merge commit will be added to master. [Modify merge commit](#)

[Like](#) [Dislike](#) [Comment](#) Oldest first Show all activity

[Elizabeth Kuzenkova](#) @E1izabeth requested review from [@E1izabeth](#) just now

[Elizabeth Kuzenkova](#) @E1izabeth assigned to [@E1izabeth](#) just now

[Elizabeth Kuzenkova](#) @E1izabeth - right now

Please, fix Use Cases for Administrator actor.

[Resolve thread](#)

Write Preview

To Do [Mark as done](#)

Assignee [Elizabeth Kuzenkova](#) @E1izabeth [Edit](#)

Reviewer [@E1izabeth](#) [Edit](#)

Milestone [None](#) [Edit](#)

Time tracking [No estimate or time spent](#) [?](#)

Labels [None](#) [Edit](#)

Lock merge request [Unlocked](#) [Edit](#)

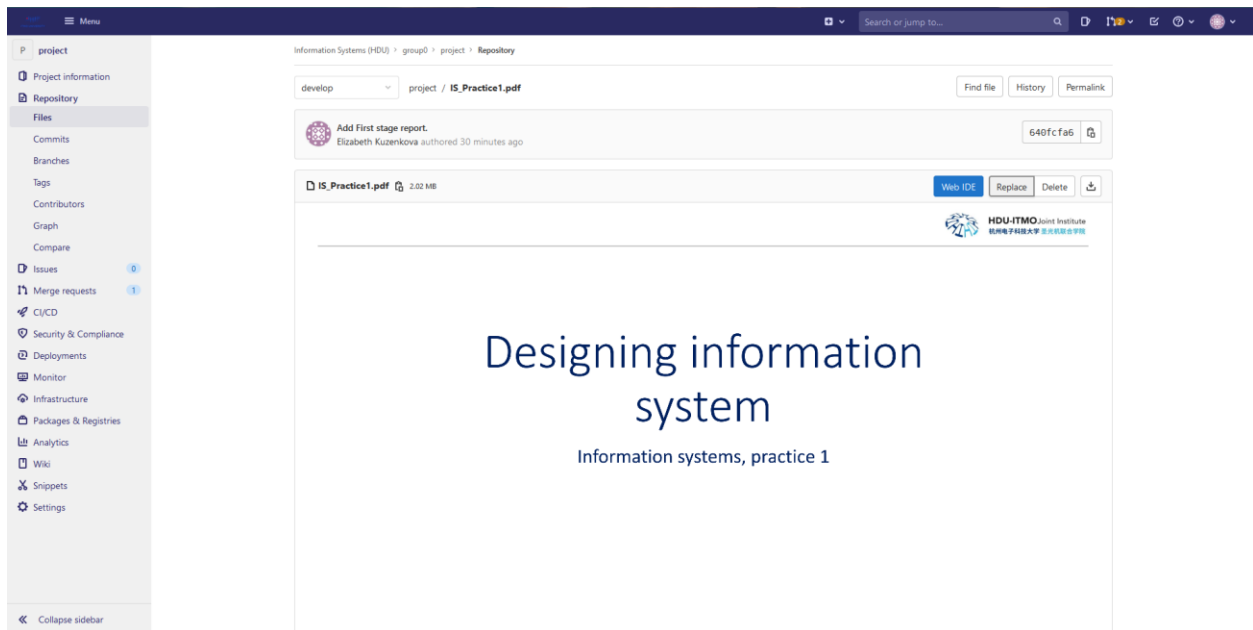
1 participant [@E1izabeth](#)

Notifications [On](#)

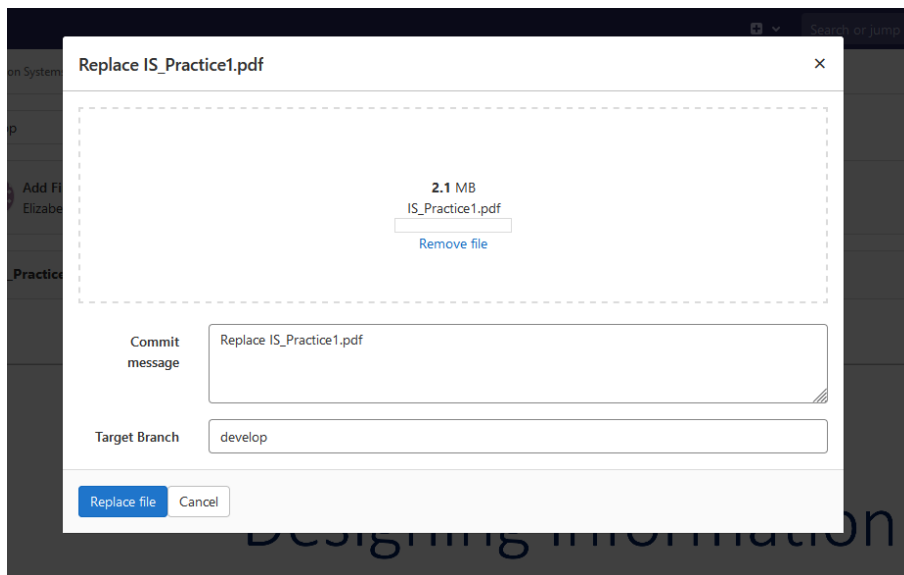
Reference: information-systems-hd... [?](#)

Source branch: `develop` [?](#)

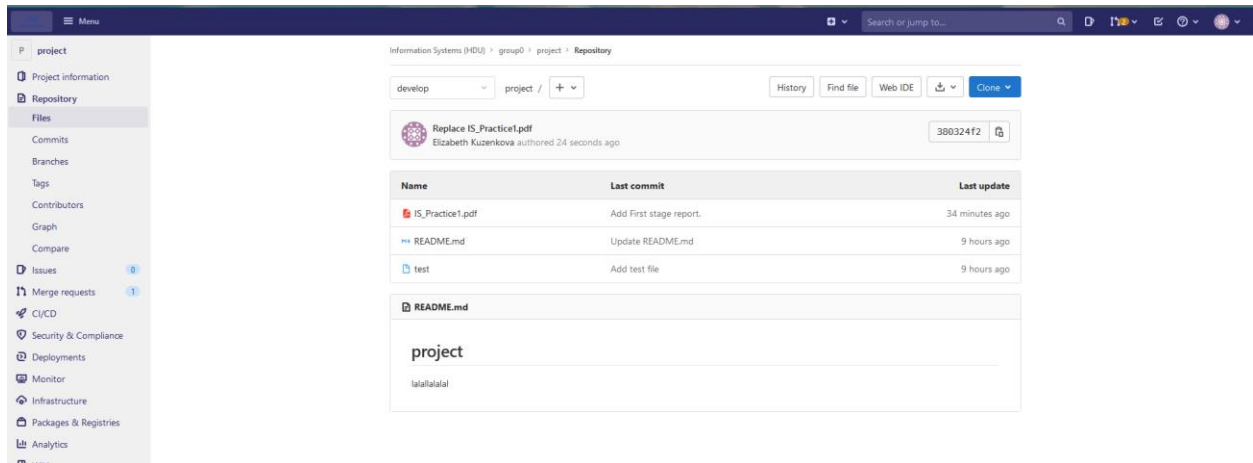
If some changes to your work needed, you should make a new commit. So, return to the «Repository» -> «Files» and choose the file where you need to make changes. Don't forget to switch branch if it's necessary! Press «Replace» button to Upload updated file. If you want to delete file, you can press «Delete» button. In both cases new commit will be automatically created.



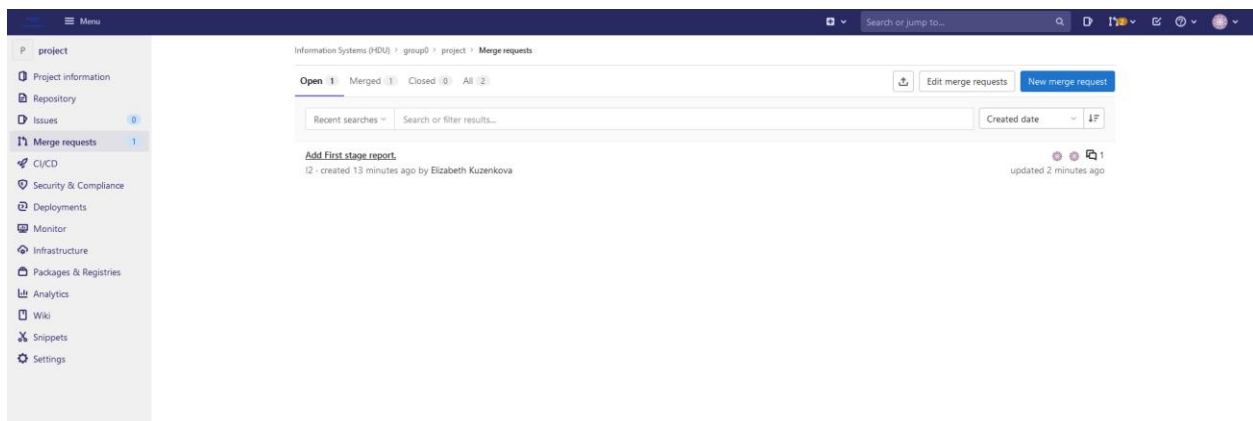
And Upload updated file in a form.



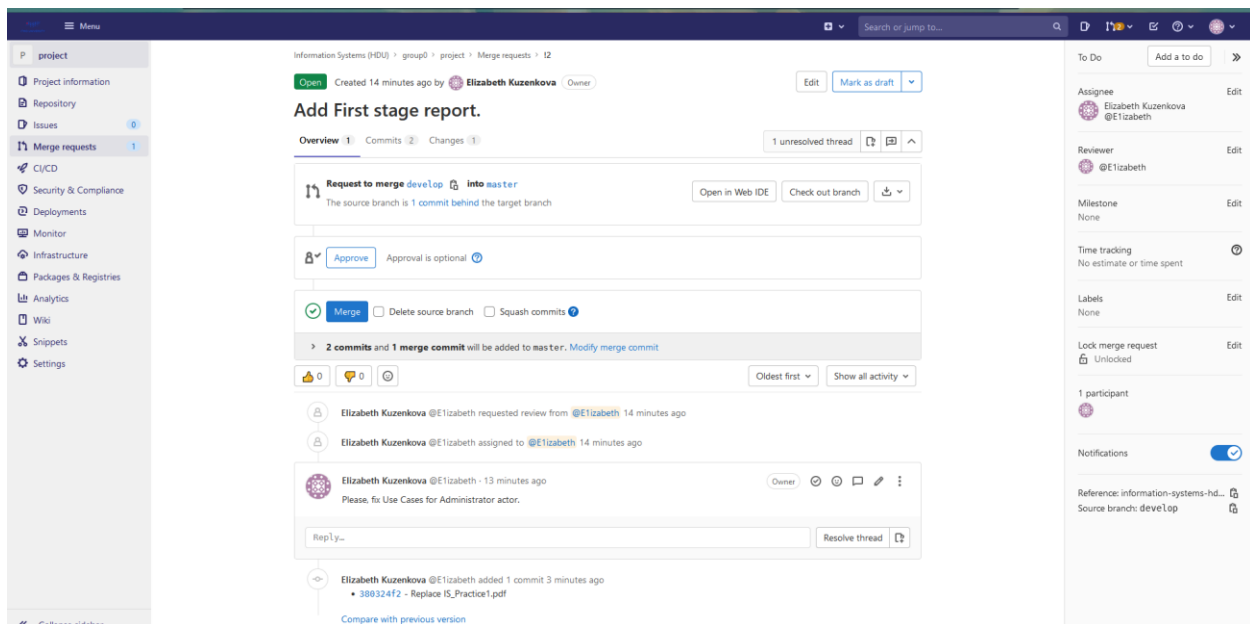
Returning to repository view we see our commit with replacing a file.



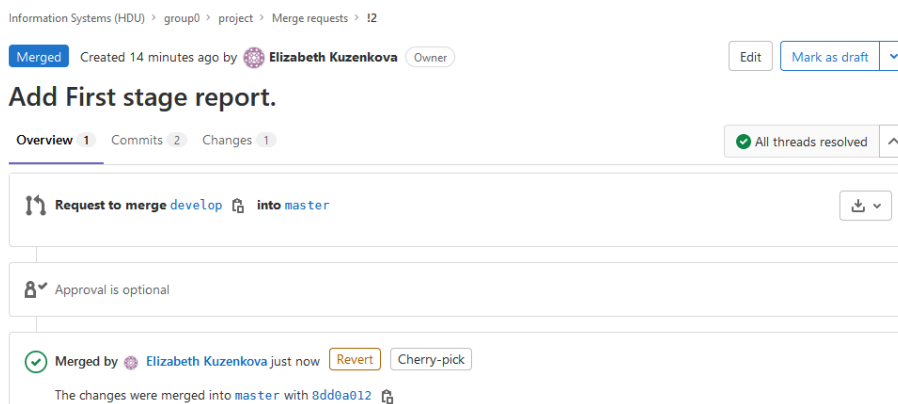
Let's see how our merge request changed. To see your merge requests press «Merge requests» button in a left menu.



Choose the merge request. Now you see that new commit was added to opened merge request. You can write a comment to a thread that you made some changes and make brief explanation or answer if needed.



If your work was accepted, then merge request will be merged.



So, state in master branch is accepted by teacher and state in develop branch is where you work.

You can edit one file together, but remember, that there could be a conflicts when you tried to commit you changes. For example, here is a conflict in merge request.

There could be a situation when GitLab couldn't propose you to choose a solution of conflict. Here is only option to merge locally.

More about how to resolve conflicts at https://docs.gitlab.com/ee/topics/git/merge_conflicts.html and https://docs.gitlab.com/ee/user/project/merge_requests/resolve_conflicts.html.

Open Created just now by **Elizabeth Kuzenkova** **Owner**

Edit **Mark as draft** **▼**

Update readme.md

Overview 0 **Commits** 2 **Changes** 1

Request to merge **develop** **into master**

The source branch is **4 commits behind** the target branch

Open in Web IDE **Check out branch** **▼**

Approve Approval is optional

Merge **There are merge conflicts** **Merge locally**

0 0

Oldest first **▼** **Show all activity** **▼**

Elizabeth Kuzenkova @E1izabeth requested review from **@E1izabeth** just now

Elizabeth Kuzenkova @E1izabeth assigned to **@E1izabeth** just now

Write **Preview**

B **I** **”** **</>** **🔗** **☰** **☰** **☰** **📎** **📄** **🔍**

Write a comment or drag your files here...

Markdown and **quick actions** are supported **📎 Attach a file**

Similar to reverting a merge request, you can opt to revert the changes directly into the target branch or create a new merge request to revert the changes.

See more at https://docs.gitlab.com/ee/user/project/merge_requests/revert_changes.html .

To do it, press on commits and choose a commit.

project

Commits

Commit 28810663 authored 5 minutes ago by **Elizabeth Kuzenkova**

Delete README.md

parent cb7ee616 **Master**

Changes 1

Showing 1 changed file with 0 additions and 3 deletions

README.md deleted 100644 -- 0

project

Some text

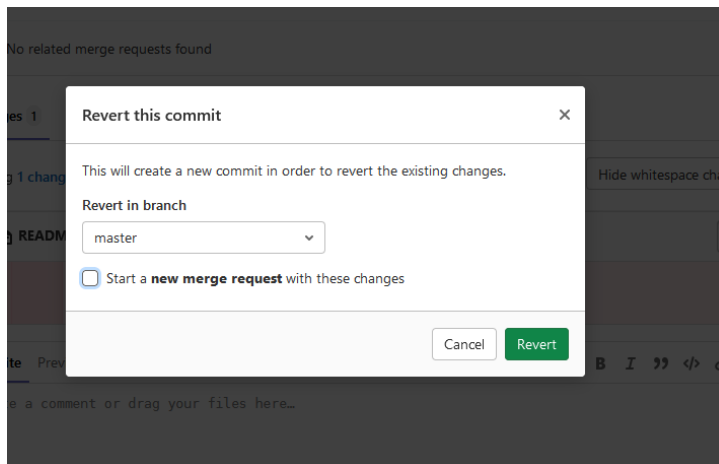
Write **Preview**

B **I** **”** **</>** **🔗** **☰** **☰** **☰** **📎** **📄** **🔍**

Write a comment or drag your files here...

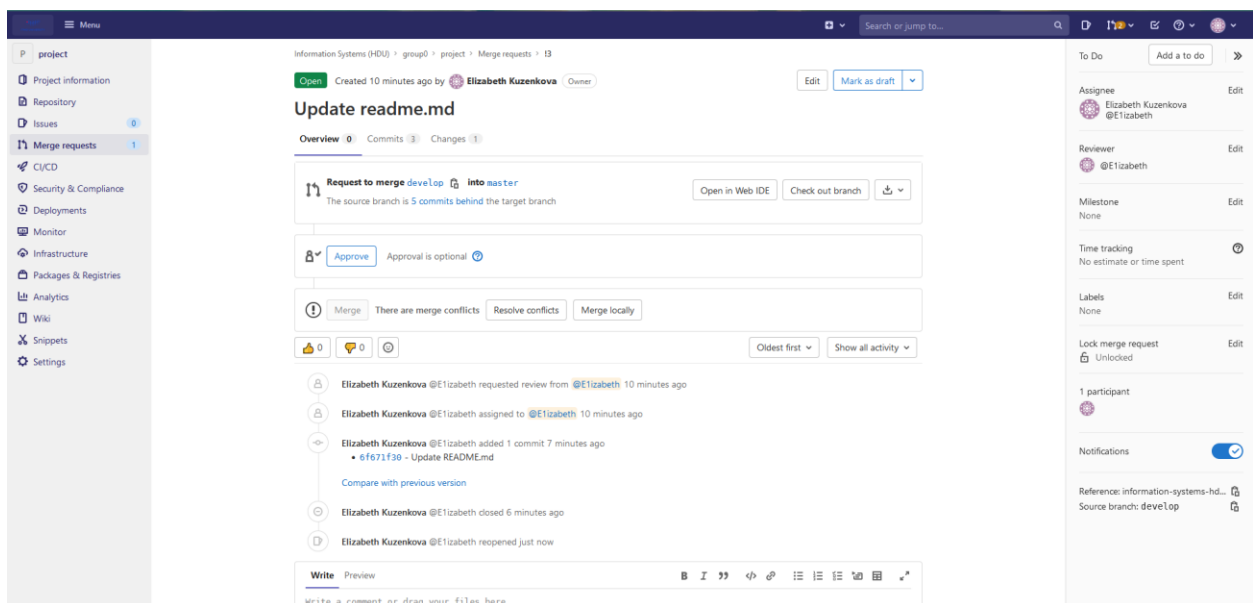
Markdown and **quick actions** are supported **📎 Attach a file**

Press «Revert» button. You will see the dialog box like presented in the next picture.

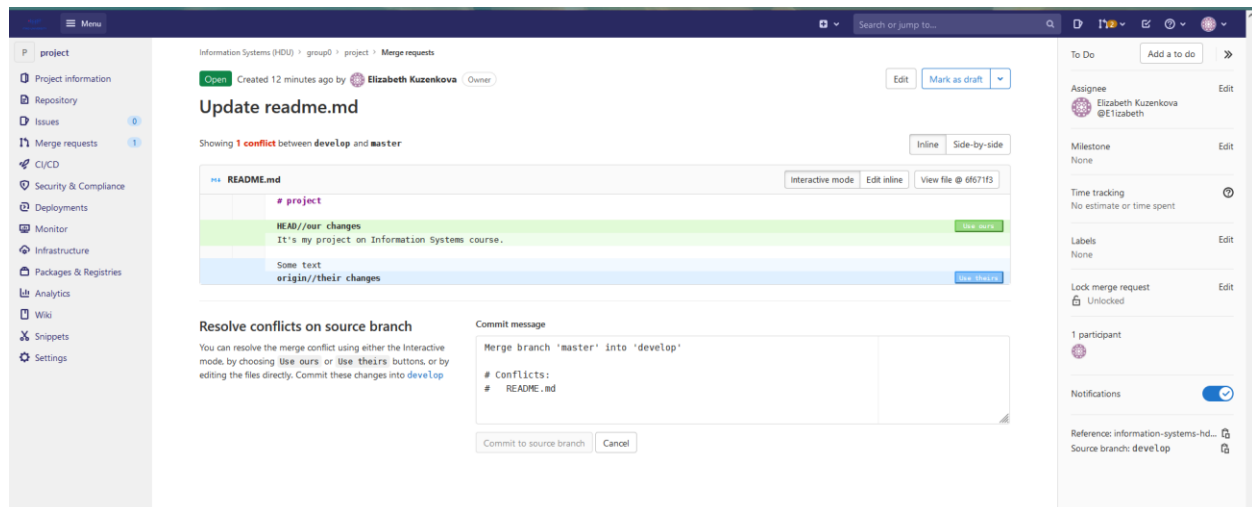


Press «Revert». After a commit is reverted, the «Revert» button is no longer available.

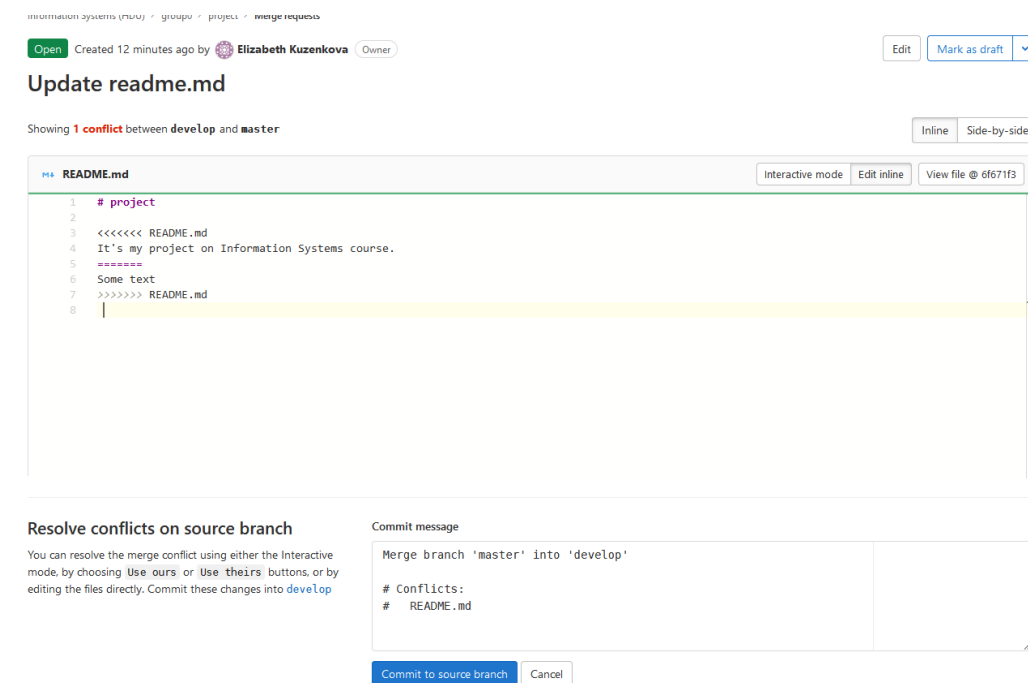
Usually you can merge pull request automatically. There is a «Resolve conflicts» button appeared.



Press «Resolve conflicts» button. You will see some editor with changes and two proposes options: *use theirs* and *use mine*. Press one of this buttons that is suitable for your situation.

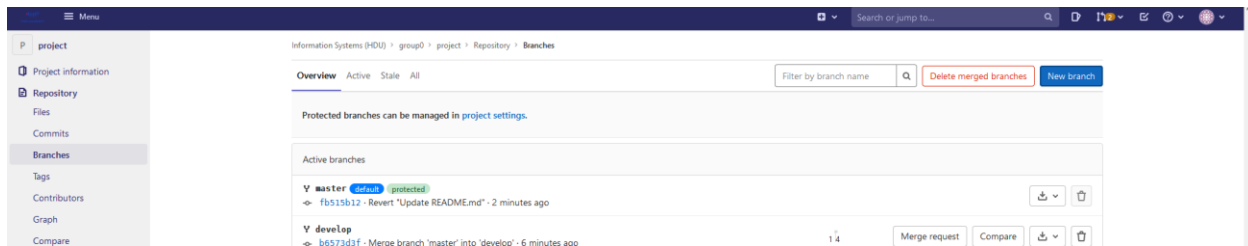


If you want to combine two variants of the file, choose «Edit inline» and made some changes.



Finally, press «Commit to source branch» button. So, commit to *develop* branch with this changes will be created. When all conflicts resolved, it will be possible to merge.

You are working in a team so you can work together in a one time but on different parts of project. Then you need different branches for work. To create new branch press «Repository» -> «Branches» button.



Press «New branch», write name of the branch in a form, choose another branch which should be the start state for the new one and press «Create branch» button.

Information Systems (HDU) > group0 > project > **New Branch**

New Branch

Branch name:

Create from:

Existing branch name, tag, or commit SHA

Another one new branch will be created.

Information Systems (HDU) > group0 > project > **Repository**

You pushed to **front-end** just now

front-end project /

Merge branch 'master' into 'develop' b6573d3f

Elizabeth Kuzenkova authored 10 minutes ago

Name	Last commit	Last update
IS_Practice1.pdf	Add First stage report.	1 hour ago
README.md	Update README.md	29 minutes ago
test	Add test file	10 hours ago

README.md

project

It's my project on Information Systems course.

Be carefully with working in different branches on the same files. Divide your subtasks between the team members to work in different branches on different files to avoid conflicts.

More about working with git and git commands:

<https://git-scm.com/book/en/v2> , <https://docs.gitlab.com/ee/topics/git/>